Forcing Reading Group — Exercise Sheet 4

The Paris–Wilkie Forcing

Remark 1. In this section we will fix a countable nonstandard $M \models \mathrm{Th}(\mathbb{N})$ and $n \in M \setminus \mathbb{N}$.

Definition 2. Let $PHP(f)$ be the formula

$$(\forall x)((\exists y \leq x)(f(y) \geq x) \vee (\exists y_1 \leq x)(\exists y_2 \leq x)(y_1 \neq y_2 \wedge f(y_1) = f(y_2))).$$

Definition 3. Let P be the set of all (standardly) finite partial injections from $\{0, \ldots, n+1\}$ to $\{0, \ldots, n\}$. For all $p, q \in P$ we define $q \leq p \Leftrightarrow q \supseteq p$.

Remark 4. Our goal is to build a more sophisticated f such that $(M, f) \models I\exists_1(f) + \neg PHP(f)$. We will build f as a union of a chain $p_0 \geq p_1 \geq p_2 \geq \ldots$ in P. We will pick the p_i's such that:

- Every pigeon $y \in \{0, \ldots, n+1\}$ is mapped somewhere.
- Every $\exists_1(f)$-formula φ satisfies $LNP(\varphi)$.

Lemma 5 (Locality of quantifier free formulas)**.** Show that if $(M, f) \models \varphi$, where f is a total union of a chain in P and φ is quantifier free $L_{PA}(f)$-formula with parameters from M, then for any $p \in P$ such that $p \subseteq f$, there is $p \subseteq q \subseteq f$ such that for all $f' \supseteq q$: $(M, f') \models \varphi$.

Definition 6. We say that $p \in P$ evaluates a closed $L_{PA}(f)$-term s in if the value of s can be evaluated recursively only on the partial structure given by p.

Exercise 7. Define a relation $\Vdash$ between P and quantifier free $L_{PA}(f)$-formulas with parameters from M as:

$$\begin{aligned} p \Vdash s = t &\Leftrightarrow (\forall q \leq p)(q \text{ evaluates } s \text{ and } t \Rightarrow q \models s = t) \\ p \Vdash s \leq t &\Leftrightarrow (\forall q \leq p)(q \text{ evaluates } s \text{ and } t \Rightarrow q \models s \leq t) \\ p \Vdash \varphi \wedge \psi &\Leftrightarrow p \Vdash \varphi \wedge p \Vdash \psi \\ p \Vdash \neg\varphi &\Leftrightarrow (\forall q \leq p)(q \nVdash \varphi). \end{aligned}$$

Show that the following equivalence holds, where f ranges over all functions which can be obtained as a total union of a chain in P:

$$p \Vdash \varphi \Leftrightarrow (\forall f \supseteq p)((M, f) \models \varphi)$$

Exercise 8. Extend the previous relation to $\exists_1(f)$-formulas with parameters from M by $p \Vdash (\exists x)\varphi(x) \Leftrightarrow (\exists m \in M)p \Vdash \varphi(m)$

Moreover, make sure that for all $p \in P$ the following holds, where f ranges over all functions which can be obtained as a total union of a chain in P:

$$(\exists q \leq p)(q \Vdash (\exists x)\varphi(x)) \Leftrightarrow (\exists f \supseteq q)(M, f) \models (\exists x)\varphi(x).$$

Exercise 9. Show that for every $\exists_1(f)$-formula φ and every $p \in P$, the set

$$\{a; \exists q \leq p : q \Vdash \varphi(a)\}$$

is definable in M.

Exercise 10. Construct an f such that $(M, f) \models I\exists_1(f) + \neg PHP(f)$.

Theorem 11. $I\exists_1(f) \nvdash PHP(f)$

Exercise 12. Consider the formula

$$\varphi_u(a) = (\forall x)(x \cdot a \leq n \to f(a + x) = a + x),$$

show that there is $(M, f) \models I\exists_1(f) + \neg LNP(\varphi_u(a))$ by constructing f as a union of a suitable chain $p_0 \geq p_1 \geq \dots$ in P.

Theorem 13. $I\exists_1(f) \nvdash LNP(\forall_1(f))$

Exercise 14. Construct an f such that $(M, f) \models I\exists_1(f) + \neg PHP(f) + \neg LNP(\varphi)$ for some $\varphi \in \forall_1(f)$.